Constructing nonabelian extensions

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...from Galois theory class: Abelian examples

$$\begin{tikzcd} \mathbb{Q}(\zeta_7) \arrow[d, dash, "6"] \\ \mathbb{Q} \end{tikzcd}$$

ζ_7 is a 7th root of unity. *e.g.* $\zeta_7 = e^{\frac{2\pi i}{7}}$.

satisfies $\frac{x^7-1}{x-1} = x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$.

the 7th roots of unity are $\zeta_7^i : 1 \leq i \leq 6$.

$\mathrm{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}) = \{\sigma_i : 1 \leq i \leq 6\}$ with $\sigma_i(\zeta_7) = \zeta_7^i$.

These satisfy $\sigma_i \circ \sigma_j(\zeta_7) = \sigma_i(\zeta_7^j) = \Big(\sigma_i(\zeta_7)\Big)^j = \zeta_7^{ij}$.

Group isomorphism exists between $\mathrm{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q})$ and the multiplicative group of the integers modulo 7.

This is a cyclic group of of order 6 generated by 3, since

$$3,\ 3^2 \equiv 2,\ 3^3 \equiv 6,\ 3^4 \equiv 4,\ 3^5 \equiv 5,\ 3^6 \equiv 1 \bmod 7.$$

As you are learning this week...

Kronecker-Weber Theorem:

Given any Galois extension $K/\mathbb{Q}$ with $\mathrm{Gal}(K/\mathbb{Q})$ abelian,

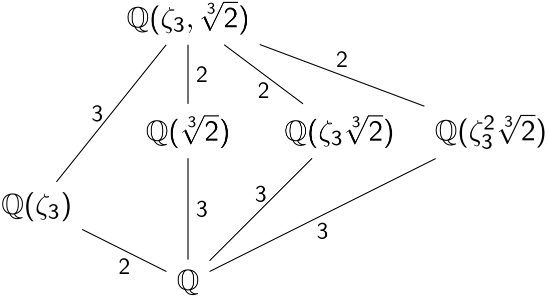
there exists an integer n such that $K \subseteq \mathbb{Q}(\zeta_n)$.

$$\begin{tikzcd}[arrows=dash]
\mathbb{Q}(\zeta_n) \arrow[d] \\
K \arrow[d] \\
\mathbb{Q}
\end{tikzcd}$$

The class examples (cyclotomic examples) you saw are a good representation of what you might see in any abelian extension.

On the other hand... nonabelian examples

In this case, we define two ring isomorphisms from $\mathbb{Q}(\zeta_3, \sqrt[3]{2})$ to itself:



σ determined by

$$\sigma(\zeta_3) = \zeta_3,$$
$$\sigma(\sqrt[3]{2}) = \zeta_3\sqrt[3]{2}.$$

τ determined by

$$\tau(\zeta_3) = \zeta_3^2 \text{ and}$$
$$\tau(\sqrt[3]{2}) = \sqrt[3]{2}.$$

$$\langle \sigma, \tau : \sigma^3 = \tau^2 = 1, \tau\sigma = \sigma^2\tau \rangle \cong D_3$$

$D_3 \cong$ the group of invertible linear functions $f_{a,b}(x) = ax + b$ modulo 3.

Alternatively, the group of invertable matrices $\begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix}$ modulo 3.

Typical class examples generalize this: $\mathbb{Q}(\zeta_n, \sqrt[n]{p})$, p prime

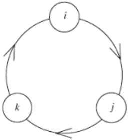
$$\begin{tikzcd}[column sep=small]
 & \mathbb{Q}(\zeta_n, \sqrt[n]{p}) \arrow[ddl, dash] \arrow[d, dash] \arrow[dr, dash] \arrow[drr, dash] & & \\
 & \mathbb{Q}(\sqrt[n]{p}) \arrow[dd, dash] & \cdots \arrow[ddl, dash] & \mathbb{Q}(\zeta_n^{n-1}\sqrt[n]{p}) \arrow[ddll, dash] \\
\mathbb{Q}(\zeta_n) \arrow[dr, dash] & & & \\
 & \mathbb{Q} & &
\end{tikzcd}$$

Similar Galois group.
Replace "3" with "n".

Note $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q} \cong U(n)$, the units modulo n, which may not be cyclic.
e.g. $U(8)$

But what about other nonabelian groups? Such as...

the group of quaternions $\{\pm 1, \pm i, \pm j, \pm k\}$?



or for any $p > 3$, the Heisenberg group modulo p?

$$H(p) = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \bmod p \right\}$$

Our mindset/orientation today

Start with an abstract p-group. Construct an extension.

In fact, construct all possible extensions that have this Galois group.

Be explicit. Produce equations.

Outline

I. Transition to characteristic p. Notation. Galois theory facts.

II. Normal Basis Theorem $\implies$ *Additive* Hilbert Theorem 90

$\implies$ Cyclic degree p extensions are Artin-Schreier.

III. Construct Heisenberg extensions.

IV. Then out of the “blue”: the Algebraic group approach.

Transition

Let p be prime,

$\mathbb{F}_p$ be the field with p elements (the integers modulo p).

The polynomial ring $\mathbb{F}_p[t]$ shares many properties with $\mathbb{Z}$.

Its quotient field, the field of rational functions

$$\mathbb{F}_p(t) = \left\{ \frac{r(t)}{s(t)} : r(t), s(t) \in \mathbb{F}_p[t], s(t) \neq 0 \right\}$$

shares properties with $\mathbb{Q}$,

and has for a long time been a source of insight and inspiration.

Some Galois theory facts

Galois extensions are separable and normal.

1. Separable: We've avoided the problem with the irreducible polynomial $x^p - t \in K[x]$ where $K = \mathbb{F}_p(t)$. Problem: Let α be a root. But then $x^p - t = x^p - \alpha^p = (x - \alpha)^p$ has only one root!

   We can "separate" the roots of the defining polynomial.
   There is a nontrivial field isomorphism that maps one root to the other.

2. Normal: We've avoided the problem with $\mathbb{Q}(\sqrt[3]{2})$. This extension doesn't contain the other two roots of $x^3 - 2$. There are isomorphisms from $\mathbb{Q}(\sqrt[3]{2})$ to other fields. *e.g.* $\mathbb{Q}(\zeta_3\sqrt[3]{2})$

   Isomorphisms are automorphisms.

Main Theorem of Galois Theory

Let L be a finite Galois extension of K with Galois group $\mathrm{Gal}(L/K)$.

1. $|\mathrm{Gal}(L/K)| = [L : K]$.
2. Bijective correspondence between subgroups of $\mathrm{Gal}(L/K)$ and intermediate fields M, $K \subseteq M \subseteq L$.

Normal Basis Theorem: Let $\mathrm{Gal}(L/K) = \{\sigma_1, \sigma_2, \ldots, \sigma_n\}$. There is an $\alpha \in L$ such that

$$\sigma_1(\alpha), \sigma_2(\alpha), \ldots, \sigma_n(\alpha)$$

is a basis for L over K.

Examples

- $\mathbb{Q}(\sqrt{2})$: The usual power basis is $\{1,\ \sqrt{2}\}$ for $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$.

  A normal basis is $\{1+\sqrt{2},\ 1-\sqrt{2}\}$. (Check linear independence.)

- $\mathbb{Q}(\zeta_5)$ where ζ_5 is a root of $x^4+x^3+x^2+x+1=0$.

  The usual power basis is $\{1,\ \zeta_5,\ \zeta_5^2,\ \zeta_5^3\}$ for $\mathbb{Q}(\zeta_5)$ over $\mathbb{Q}$.

  A normal basis is $\{1+\zeta_5,\ 1+\zeta_5^2,\ 1+\zeta_5^3,\ 1+\zeta_5^4\}$.

  (Again, check linear independence.)

Artin-Schreier extensions

Let $K = \mathbb{F}_p(t)$, and suppose $x^p - x = \beta \in K$ is irreducible with root α.

Then $\alpha + 1$ is another root. Indeed, $\{\alpha + i : i \in \mathbb{F}_p\}$ is the set of roots.

$K(\alpha)/K$ is Galois with $\mathrm{Gal}(K(\alpha)/K) = \langle \sigma \rangle \cong C_p$, $\sigma(\alpha) = \alpha + 1$.

We want to prove the converse.

Stepping stone: Additive Hilbert Theorem 90

Let L/K be a Galois extension with $\mathrm{Gal}(L/K) \cong C_p$ generated by σ.
Thus $\sigma^p = 1$, the identity.

The trace $\mathrm{Tr} = 1 + \sigma + \sigma^2 + \cdots + \sigma^{p-1}$ is a K-linear map that we can apply to elements of L: Given $\mu \in L$, $\mathrm{Tr}(\mu) = \mu + \sigma(\mu) + \cdots + \sigma^{p-1}(\mu)$.

Suppose $\ell \in L$ with $\mathrm{Tr}(\ell) = 0$. There is an $\alpha \in L$ such that $\ell = \sigma(\alpha) - \alpha$.

We think of $\sigma(\alpha) - \alpha$ as the result of the K-linear map $(\sigma - 1)$ acting on the "vector" α, and write $\ell = (\sigma - 1)\alpha$.

First, let's verify: If $\ell = (\sigma - 1)\alpha$, then $\mathrm{Tr}(\ell) = 0$.

Proof of Additive Hilbert Theorem 90

Preparation: Using the Normal Basis Theorem, there is an $\mu \in L$ such that

$$\{\mu, \sigma(\mu), \sigma^2(\mu), \dots, \sigma^{p-1}(\mu)\}$$

is a K-basis for L.

Notice that: $\mathrm{Tr}(\mu) = \mu + \sigma(\mu) + \cdots + \sigma^{p-1}(\mu) \neq 0$.

Change basis: For $1 \leq i \leq p-1$, replace $\sigma^i(\mu)$ with $\sigma^i(\mu) - \mu$.

Adjust notation: $\sigma^i(\mu) - \mu = (\sigma^i - 1)\mu$.

Think: $(\sigma^i - 1)$ is a K-linear function acting on the vector μ.

New basis: $\{\mu, (\sigma - 1)\mu, (\sigma^2 - 1)\mu, \dots, (\sigma^{p-1} - 1)\mu\}$.

Notice that: $\mathrm{Tr}((\sigma^i - 1)\mu) = 0$ for $1 \leq i \leq p-1$, because

$$\{\sigma^i(\mu), \sigma^{i+1}(\mu), \sigma^{i+2}(\mu), \dots, \sigma^{i+p-1}(\mu)\} = \{\mu, \sigma(\mu), \sigma^2(\mu), \dots, \sigma^{p-1}(\mu)\},$$

and thus $\mathrm{Tr}(\sigma^i(\mu)) = \mathrm{Tr}(\mu)$.

We are ready!

Suppose $\ell \in L$ and $\mathrm{Tr}(\ell) = 0$. Since $\ell \in L$, there are $a_i \in K$ such that

$$\ell = a_0\mu + a_1(\sigma - 1)\mu + a_2(\sigma^2 - 1)\mu + \cdots + a_{n-1}(\sigma^{n-1} - 1)\mu.$$

Apply the trace to both sides.

Since $\mathrm{Tr}(\ell) = 0$ and $\mathrm{Tr}((\sigma^i - 1)\mu) = 0$ for $1 \leq i \leq p - 1$,

$$0 = \mathrm{Tr}(a_0\mu) = a_0\mathrm{Tr}(\mu) \text{ with } \mathrm{Tr}(\mu) \neq 0.$$

This means $a_0 = 0$.

And thus that ℓ is a sum of terms $a_i(\sigma^i - 1)\mu$, $i \geq 1$.

Factor $(\sigma - 1)$ out of each term.

We have proven $\ell = (\sigma - 1)\alpha$ for some $\alpha \in L$. Done!

Proving C_p-extensions are Artin-Schreier

Suppose L/K is Galois with $\mathrm{Gal}(L/K) \cong C_p$ generated by σ.

Since the characteristic of $K = \mathbb{F}_p(t)$ is p,

$$\mathrm{Tr}(1) = 1 + \sigma(1) + \cdots + \sigma^{p-1}(1) = p = 0.$$

Using additive Hilbert Thm 90, there is an $\alpha \in L$ such that $1 = (\sigma - 1)\alpha$.

Rewrite as $\sigma(\alpha) = \alpha + 1$. Since $\sigma(\alpha) \neq \alpha$, $K \subsetneq K(\alpha) \subseteq L$ where because $[L : K] = p$, we have $L = K(\alpha)$. Induction yields $\sigma^i(\alpha) = \alpha + i$.

Recall Fermat's little Theorem: $0, 1, \ldots, p-1$ solve $x^p \equiv x \bmod p$.

So $x^p - x$ factors as $\prod_{i \in \mathbb{F}_p}(x - i)$, which means that

$$\alpha^p - \alpha = \prod_{i \in \mathbb{F}_p}(\alpha - i) = \prod_{i=0}^{p-1} \sigma^i(\alpha) \in K.$$

i.e. α is a root of the Artin-Schreier equation $x^p - x = \beta$ for some $\beta \in K$.

We are ready to start constructing Heisenberg extensions.

actually... Galois extensions for any p-group.

Start with the group

$$H(p): \quad \sigma_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

These elements satisfy $\sigma_i^p = 1$. The center is $\langle \sigma_3 \rangle$. And $\sigma_2\sigma_1 = \sigma_1\sigma_2\sigma_3$.

Importantly, $H(p)/\langle \sigma_3 \rangle \cong C_p \times C_p$ is elementary abelian, and we know how elementary abelian extensions are constructed!

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Begin: Suppose N/K is a Galois extension with $\mathrm{Gal}(N/K) = \langle \sigma_1, \sigma_2, \sigma_3 \rangle$, and let $M = N^{\sigma_3}$ be the fixed field of $\langle \sigma_3 \rangle$.

Then since $\mathrm{Gal}(M/K) = \dfrac{\langle \sigma_1, \sigma_2, \sigma_3 \rangle}{\langle \sigma_3 \rangle} \cong C_p \times C_p$,

the extension M/K is built out of two Artin-Schreier C_p-extensions.

Emerging picture

$$\begin{tikzcd}
 & N \arrow[d, dash, "\sigma_3"] & \\
 & M = K(\alpha_1, \alpha_2) \arrow[dl, dash, "\sigma_2"'] \arrow[dr, dash, "\sigma_1"] & \\
L = K(\alpha_1) \arrow[dr, dash] & & K(\alpha_2) \arrow[dl, dash] \\
 & K = \mathbb{F}_p(t) &
\end{tikzcd}$$

$M = K(\alpha_1, \alpha_2)$ with

α_i satisfying $x^p - x = \beta_i \in K$,

and $\sigma_i(\alpha_j) = \alpha_j + \delta_{ij}$, $i, j \in \{1, 2\}$,

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases},$$

Kronecker delta.

Since N/M is a C_p-extension, $N = M(\alpha_3)$ for some $\alpha_3^p - \alpha_3 = \mu \in M$.
But M has lots of C_p-extensions. When do we have a $H(p)$-extension?

This is an *embedding problem*.

Recall: α_1, α_2 satisfying $\sigma_3(\alpha_j) = \alpha_j$, solving $x^p - x = \beta_j \in K$. And

$$\sigma_i(\alpha_j) = \alpha_j + \delta_{ij} \text{ for } i, j \in \{1, 2\}.$$

Now since σ_3 is in the center, $\langle \sigma_2, \sigma_3 \rangle \cong C_p \times C_p$. Thus N is the composite of two Artin-Schreier extensions of L. Namely, $N = L(\alpha_2, \alpha_3)$ with $\alpha_3^p - \alpha_3 \in L$ and $\sigma_2(\alpha_3) = \alpha_3$

$$\begin{tikzcd}[column sep=small]
 & N \arrow[d, dash, "\sigma_3"] & \\
 & M = K(\alpha_1, \alpha_2) \arrow[dl, dash, "\sigma_2"'] \arrow[dr, dash, "\sigma_1"] & \\
L = K(\alpha_1) \arrow[dr, dash] & & K(\alpha_2) \arrow[dl, dash] \\
 & K = \mathbb{F}_p(t) &
\end{tikzcd}$$

In summary,
$\sigma_i(\alpha_j) = \alpha_j + \delta_{ij}$ for $i, j \in \{1, 2, 3\}$,
except one pair: $(i, j) = (1, 3)$.

For that one pair, all we can say is that $\sigma_1(\alpha_3) \in N$.

So, $\sigma_1(\alpha_3) - \alpha_3 = A$,
for some $A \in N$.

Goal: Determine A.

Determine A

Recall $A = (\sigma_1 - 1)\alpha_3 \in N = K(\alpha_1, \alpha_2, \alpha_3)$.

I. Since $(\sigma_3 - 1)\alpha_3 = 1$ and the K-linear maps $(\sigma_1 - 1)$ and $(\sigma_3 - 1)$ commute, $(\sigma_3 - 1)A = (\sigma_1 - 1)(\sigma_3 - 1)\alpha_3 = 0$.

Thus $A \in M = K(\alpha_1, \alpha_2)$.

II. Use $\sigma_2\sigma_1 = \sigma_1\sigma_2\sigma_3$. Since

$$\sigma_2\sigma_1(\alpha_3) = \sigma_2(\alpha_3 + A) = \alpha_3 + \sigma_2(A), \text{ and}$$
$$\sigma_1\sigma_2\sigma_3(\alpha_3) = \sigma_1\sigma_2(\alpha_3 + 1) = \sigma_1(\alpha_3 + 1) = \alpha_3 + A + 1,$$

We find that $\sigma_2(A) = A + 1$.

Since $A \in M$ and $\sigma_2(\alpha_2) = \alpha_2 + 1$, we find that σ_2 fixes $A - \alpha_2 \in M$.

Thus $A - \alpha_2 \in L = K(\alpha_1)$.

Determining A

We have shown $A = (\sigma_1 - 1)\alpha_3$ satisfies $A - \alpha_2 \in L = K(\alpha_1)$.

III. Using $\sigma_1(\alpha_3) = \alpha_3 + A$, we find

$$\begin{aligned}
\sigma_1(\alpha_3) &= \alpha_3 + A, \\
\sigma_1^2(\alpha_3) &= \alpha_3 + A + \sigma_1(A), \\
\sigma_1^3(\alpha_3) &= \alpha_3 + A + \sigma_1(A) + \sigma_1^2(A), \\
&\vdots \\
\sigma_1^p(\alpha_3) &= \alpha_3 + A + \sigma_1(A) + \sigma_1^2(A) + \cdots + \sigma_1^{p-1}(A).
\end{aligned}$$

Thus $\mathrm{Tr}_{\langle\sigma_1\rangle}(A) = 0$.

Meanwhile σ_1 fixes α_2. Thus $\mathrm{Tr}_{\langle\sigma_1\rangle}(\alpha_2) = p = 0$ as well.

Therefore $A - \alpha_2 \in L = K(\alpha_1)$ and $\mathrm{Tr}_{\langle\sigma_1\rangle}(A - \alpha_2) = 0$. Additive Hilbert Theorem 90 kicks in and there is an $\ell \in L$ such that

$$A - \alpha_2 = (\sigma_1 - 1)\ell.$$

Determining A

Recall that $L(\alpha_3)/L$ is a C_p-extension, that $\alpha_3^p - \alpha_3 \in L$ and that $(\sigma_3 - 1)\alpha_3 = 1$, $(\sigma_2 - 1)\alpha_3 = 0$.

Furthermore, $A = (\sigma_1 - 1)\alpha_3$ and $A - \alpha_2 = (\sigma_1 - 1)\ell$ with $\ell \in L$.

Set $\alpha_3' = \alpha_3 - \ell$. Then since L is fixed by σ_2, σ_3, we have $(\sigma_3 - 1)\alpha_3' = 1$, $(\sigma_2 - 1)\alpha_3' = 0$. Additionally,

$$(\sigma_1 - 1)\alpha_3' = (\sigma_1 - 1)(\alpha_3 - \ell) = A - (\sigma_1 - 1)\ell = \alpha_2.$$

So, we replace α_3 with α_3' and then relabel.

Without loss of generality, $(\sigma_1 - 1)\alpha_3 = \alpha_2$.

What does this tell us about the element $\mu \in L$ such that $\alpha_3^p - \alpha_3 = \mu$?

$$\sigma_1(\mu) = (\sigma_1(\alpha_3))^p - \sigma_1(\alpha_3) = (\alpha_3 + \alpha_2)^p - (\alpha_3 + \alpha_2) = \mu + \beta_2$$

Determining $\mu = \alpha_3^p - \alpha_3$

We have $\sigma_1(\mu) = \mu + \beta_2$. Thus $(\sigma_1 - 1)\mu = \beta_2$.

Since $(\sigma_1 - 1)(\beta_2\alpha_1) = \beta_2$ as well.

We find that because $\mu - \beta_2\alpha_1 \in L = K(\alpha_1)$ is fixed by σ_1,

$$\mu - \beta_2\alpha_1 \in K.$$

In other words,

$$\alpha_3^p - \alpha_3 = \alpha_1\beta_2 + \beta_3 \text{ for some } \beta_3 \in K.$$

Summary

Let $K = \mathbb{F}_p(t)$. If N/K is a Galois extension with $\mathrm{Gal}(N/K) \cong H(p)$, then $N = K(\alpha_1, \alpha_2, \alpha_3)$ with

$$\begin{aligned} \alpha_1^p - \alpha_1 &= \beta_1, \\ \alpha_2^p - \alpha_2 &= \beta_2, \\ \alpha_3^p - \alpha_3 &= \alpha_1\beta_2 + \beta_3. \end{aligned}$$

And the converse holds, assuming the polynomial equations are irreducible.

Quaternion answer.

Quaternion extensions of $K = \mathbb{F}_2(t)$: $N = K(\alpha_1, \alpha_2, \alpha_3)$ with

$$\begin{aligned}
\alpha_1^2 - \alpha_1 &= \beta_1, \\
\alpha_2^2 - \alpha_2 &= \beta_2, \\
\alpha_3^2 - \alpha_3 &= \alpha_1\beta_1 + \alpha_1\beta_2 + \alpha_2\beta_2 + \beta_3.
\end{aligned}$$

For more exercises and an answer-key!

Grant Moles, *Non-Abelian extensions of degree p^3 and p^4 in characteristic $p > 2$.* Finite Fields Appl. **96** (2024).

Algebraic group approach

Interestingly (and for me, surprisingly!), the Heisenberg system of equations

$$\begin{aligned}
\alpha_1^p - \alpha_1 &= \beta_1, \\
\alpha_2^p - \alpha_2 &= \beta_2, \\
\alpha_3^p - \alpha_3 &= \alpha_1\beta_2 + \beta_3.
\end{aligned}$$

amounts to the single matrix equation

$$\begin{pmatrix} 1 & \alpha_1^p & \alpha_3^p \\ 0 & 1 & \alpha_2^p \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & \alpha_1 & \alpha_3 \\ 0 & 1 & \alpha_2 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & \beta_1 & \beta_3 \\ 0 & 1 & \beta_2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$H(R) = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in R \right\}$$ is a group regardless of ring R.

In this talk, $\mathrm{char}(R) = p$.

Since $\alpha_1, \alpha_2, \alpha_3 \in N$, the that matrix equation exists in the group $H(N)$.

The Galois group is the $\mathbb{F}_p$-points $H(\mathbb{F}_p)$ of this large group.

Galois action is described with a matrix equation (omitted).

The point is that this is a really nice set-up:
Whenever a Galois group is the $\mathbb{F}_p$-points of such an *Algebraic group*, this sort of thing works.

(Pointed out to me by Kevin Keating, Univ Florida)

Down a rabbit-hole...

You might now wonder whether all p-groups can be realized as the $\mathbb{F}_p$-points an algebraic group. And you aren't the first.

David Speyer (Univ Michigan) believes D_8 (order 16) to be the smallest counter-example

`mathoverflow.net/questions/262745/`

9 years. Still open.

Of course...

I determined the Artin-Schreier equations that produce D_8-extensions in characteristic 2.

Yes, reverse engineering Keating's observation, yields a binary operation:

$$\begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} * \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} x_0 + y_0 \\ x_1 + y_1 \\ x_2 + y_2 + x_0 y_1 + x_1 y_1 \\ x_3 + y_3 + D + E + F \end{bmatrix}$$

where $D = x_1^3 y_1 + x_1 y_1^3 + x_1 x_2 y_1 + x_1 y_1 y_2 + x_2 y_2$, $E = x_0 x_1 y_1^2 + x_0 x_2 y_1$, and $F = x_0(y_1 + y_2 + y_1 y_2 + y_0 y_1^2)$.

But no, this doesn't produce an algebraic group (Group operation defined by polynomials).

It's not associative!

Thanks!